

HAHNENKAMM EHRENBACHHOEHE, Austria

WMO: 111350

Lat: **47.418N**Lon: **12.359E**Elev: **1764**StdP: **81.86**Time Zone: **1.00 (EUC)**Period: **94-10**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.1	-13.4	-21.7	0.7	-4.9	-19.1	0.9	-5.6	10.1	-4.2	8.4	-3.6	1.7	270	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	20.1	13.4	18.7	12.9	17.4	12.2	14.7	18.4	13.7	17.3	12.9	16.6	2.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	11.9	16.4	12.2	11.0	15.3	11.3	10.3	14.6	47.5	18.6	44.5	17.4	42.1	16.6	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.8	5.7	DB	-17.0	22.5	2.8	2.2	-19.1	24.1	-20.7	25.4	-22.3	26.6	-24.3	28.2	(4)
(5)			WB	-17.3	16.2	2.7	1.5	-19.2	17.2	-20.8	18.1	-22.3	18.9	-24.2	20.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-3.3	-4.0	-2.0	1.7	6.7	9.4	11.2	10.9	8.0	5.7	0.5	-2.1	(6)	
	DBStd	7.31	4.87	5.29	5.32	4.82	4.59	4.68	4.18	3.88	4.48	4.98	5.55	5.25	(7)	
	HDD10.0	2584	412	391	373	251	123	67	37	36	88	144	287	374	(8)	
	HDD18.3	5378	671	624	632	499	362	269	222	229	311	392	535	632	(9)	
	CDD10.0	250	0	0	0	2	20	49	75	66	27	11	1	0	(10)	
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0	(11)	
	CDH23.3	3	0	0	0	0	0	0	3	0	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.5	2.7	2.7	2.7	2.6	2.6	2.5	2.5	2.4	2.2	2.4	2.7	2.5	(14)	
(15) Precipitation	PrecAvg	1309	66	61	84	83	134	168	180	175	123	95	76	66	(15)	
	PrecMax	1511	183	181	214	167	197	253	280	261	234	184	152	159	(16)	
	PrecMin	1077	7	5	28	5	33	96	57	108	57	22	0	8	(17)	
	PrecStd	110	40	39	43	40	46	40	51	43	38	39	37	36	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	9.7	10.1	13.5	18.9	20.8	22.9	20.9	18.2	16.7	13.5	9.8	(19)	
		MCWB	1.8	2.6	3.6	6.6	11.2	13.3	14.4	14.1	11.5	9.3	6.2	2.4	(20)	
	2%	DB	7.1	7.2	8.1	11.5	16.6	19.2	20.6	19.1	16.5	14.7	11.3	7.7	(21)	
		MCWB	0.4	0.5	2.8	5.6	10.2	12.9	14.2	13.2	10.6	8.1	5.0	1.9	(22)	
	5%	DB	5.2	5.2	6.7	9.7	15.0	17.7	19.0	17.7	15.1	13.3	9.4	6.1	(23)	
		MCWB	0.2	-0.3	1.9	4.6	9.3	12.2	13.5	12.8	10.2	7.5	3.9	0.6	(24)	
	10%	DB	3.4	3.3	5.3	8.2	13.2	16.2	17.4	16.4	13.8	12.0	8.0	4.7	(25)	
		MCWB	-0.7	-0.8	1.2	3.6	8.3	11.2	12.8	12.2	9.6	7.2	3.0	0.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.3	4.7	7.6	12.3	15.1	16.8	15.2	13.1	10.7	7.4	4.1	(27)	
		MCDB	7.1	8.2	8.0	11.9	17.0	19.0	19.7	18.4	15.7	13.8	12.3	8.1	(28)	
	2%	WB	1.7	1.5	3.7	6.1	10.9	13.7	15.3	14.1	11.9	9.6	5.6	2.5	(29)	
		MCDB	5.6	5.5	6.7	11.1	15.2	17.5	19.0	17.4	14.7	12.8	9.8	5.7	(30)	
	5%	WB	0.5	0.6	2.8	5.0	9.7	12.8	14.1	13.4	11.0	8.6	4.5	1.6	(31)	
		MCDB	4.1	4.1	5.6	8.9	13.7	16.6	17.8	16.7	14.0	12.0	8.1	4.9	(32)	
	10%	WB	-0.5	-0.4	1.8	4.0	8.8	11.9	13.1	12.6	10.1	7.7	3.5	0.7	(33)	
		MCDB	2.8	2.7	4.5	7.5	12.5	15.5	16.9	15.8	13.0	11.1	7.2	4.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.8	4.9	4.8	5.2	5.2	5.6	4.9	4.6	5.0	4.7	4.6	(35)	
		MCDBR	6.7	6.1	5.9	5.6	6.5	6.6	7.5	6.1	5.8	5.8	5.4	5.8	(36)	
	5% WB	MCWBR	3.9	3.7	3.4	3.5	4.0	4.2	4.6	4.2	4.2	3.8	3.1	3.6	(37)	
		MCWBR	6.3	5.9	5.6	5.4	5.9	6.2	6.8	5.7	5.4	5.2	5.4	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.235	0.255	0.287	0.316	0.328	0.334	0.334	0.330	0.305	0.283	0.259	0.235	(40)	
		taud	2.458	2.409	2.404	2.424	2.439	2.448	2.439	2.471	2.528	2.572	2.568	2.506	(41)	
	Edn at Noon		886	931	943	940	935	927	922	915	912	888	850	847	(42)	
		Edh at Noon	62	82	98	106	109	109	108	101	86	70	57	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.30	2.24	3.23	4.12	4.69	5.20	5.27	4.69	3.44	2.40	1.36	1.06	(44)	
	RadStd		0.16	0.28	0.38	0.52	0.56	0.54	0.51	0.56	0.45	0.26	0.22	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)		+0.42	N/A	N/A	N/A	+0.36	+0.36	N/A	N/A	+62	+29

Nomenclature: See separate page